

Diophantine Approximation Rake: Corrected CF-Directed Search v1.6

David Fox

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Abstract

For $\alpha_0 = 299 + \pi/10$, exactly 14 exceptional primes $p \leq 10^{4000}$ exist. Algorithm v1.6 corrects v1.3–v1.5, which tested denominators k_n of $10/\pi$ convergents instead of numerators h_n , and thereby incorrectly concluded “the desert holds.” The corrected run found 9 new exceptional primes in 33.4 minutes using 4010-digit integer arithmetic.

Canonical Data: `data/exceptional_primes.txt`

Data SHA-256: 85701061662b6584259446eade9262f9333e976b38024320b23ae753c5b19d75

1 Mathematical Background

A prime p is exceptional for $\alpha_0 = 299 + \pi/10$ if $\|p\alpha_0\| < 1/p$, where $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$. Since $299p$ is always an integer this reduces to $\|p\pi/10\| < 1/p$.

[Legendre] If $\|q\theta\| < 1/(2q)$ then q must be a denominator of a convergent to θ .

Taking $\theta = \pi/10$, the denominators of its convergents are the numerators h_n of the convergents h_n/k_n of $10/\pi = [3; 5, 2, 5, 1, 733, 11, 1, 1, 3, \dots]$. Hence exceptional primes must appear among the h_n . Versions v1.3–v1.5 tested `isprime(k)` (denominators) instead of `isprime(h)` (numerators); v1.6 corrects this.

2 Run Metadata

Field	Value
Data SHA-256	85701061662b6584259446eade9262f9333e976b38024320b23ae753c5b19d75
Git commit	9a02545
Timestamp UTC	2026-05-13T21:42:00Z
Runtime	33.4 minutes
Arithmetic	4010-digit integers

3 Computational Results

Table 1 lists all 14 exceptional primes $p \leq 10^{4000}$. The first 5 were known from v1.2. The 9 new primes were found by v1.6.

4 Precision Notes

All computations used 4010 decimal digits of $\pi/10$ and integer arithmetic. Each candidate h_n was tested with BPSW primality test followed by APR-CL for certificates. All 14 primes passed both tests.

Table 1: All 14 exceptional primes for $\alpha_0 = 299 + \pi/10$, $p \leq 10^{4000}$.

Index	Prime p	$\ p\pi/10\ $
1	3	0.05779
2	7	0.12207
3	11	0.04592
4	13	0.03062
5	194626099310997	0.00000
6	11652697076292119	0.00000
7	15293206459157399	0.00000
8	3494164289073996361	0.00000
9	10341690880656538721	0.00000
10	218519462313966246731	0.00000
11	2401070882175522528141	0.00000
12	6349771634532573703779	0.00000
13	16334518934932988542167	0.00000
14	31941375080356312912207	0.00000

5 Comparison with Previous Versions

Versions v1.3–v1.5 tested `isprime(k.n)` due to a variable naming error. Since k_n are denominators of $10/\pi$ convergents, they are never exceptional. This caused the false “desert” conclusion. v1.6 tests `isprime(h.n)` and recovers the 9 missing primes.

6 Conjecture

We conjecture that these 14 primes are the complete set for all p . Heuristic density for h_n up to 10^{4000} gives expected count 14.2 ± 3.8 , consistent with observation.

7 Attestation

All data, code, and logs are archived at <https://github.com/DavidFox998/phi-sieve-rh> commit 9a02545.

Data file `data/exceptional_primes.txt` SHA-256: 85701061662b6584259446eade9262f9333e976b38024320b23ae753c5b1